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Surface EMG Biofeedback for Hand Recovery After Stroke: A Randomized Trial

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Abstract

Results show that continuous sEMG biofeedback improved hand function in stroke patients who followed the grip rehabilitation protocol. Data are in Dataset A.

1. Introduction

Biofeedback can guide effort during training. Evidence for its effect on hand function is limited.

2. Methodology

Stroke patients were randomised to the grip rehabilitation protocol. Continuous sEMG was recorded for all sessions.

3. Results

Hand function scores improved significantly in stroke patients after the grip rehabilitation protocol. Muscle activation increased significantly during grip tasks measured with continuous sEMG. Grip strength also improved in stroke patients.

4. Discussion and Limitations

Limitations include short follow-up. The effect in other diseases has not been studied.

5. Conclusion

The grip rehabilitation protocol with continuous sEMG is a promising option.

References

[1] Fictional A. et al. Synthetic reference. 2024.